GQ-TSR: Guarded Query Teacher–Student Retrieval for CBCT-to-Clinical Record Prediction

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Abstract. Predicting a structured dental record from cone beam computed tomography (CBCT) is difficult because paired records are scarce and several clinical fields are only partly observable from the image. We present GQ-TSR, a Guarded Query Teacher Student Retrieval framework that predicts seven clinical record fields from CBCT alone while using unpaired scans during training. Training follows three stages. First, image only pretraining uses all training CBCT volumes to initialize the three dimensional encoder. Then, supervised learning uses paired volumes and records to learn global and local visual representations together with structured field semantics. Finally, a guarded EMA teacher and student stage incorporates unpaired scans while retaining the supervised solution when semi supervised training is not beneficial. The selected models retrieve each field from labeled reference records, and their predictions are combined across folds. On the STS 2026 Challenge public validation set, GQ-TSR achieved a Weighted Score of 0.26, a field completion rate of 1.00, and no missing predictions.

Keywords: Dental CBCT · Semi-supervised learning · Record retrieval

1 Introduction

Cone beam computed tomography (CBCT) depicts teeth and surrounding structures in three dimensions [1], while structured dental records describe symptoms, examinations, diagnoses, treatments, and advice. MMDental links these modalities through paired CBCT volumes and expert records [2]. Predicting such records remains difficult because small findings require both global context and local detail, while complaints, history, and treatment decisions are only partly observable from the image. Dental vision language research supports structured image and text reasoning [3], and three dimensional models show the value of combining global and local evidence [4]. Adaptation through self supervision also improves image and report alignment when institutional data are limited [5]. Nevertheless, scarce paired records and the gap between voxels and clinical language make direct generation vulnerable to overfitting and unsupported statements [6].

Semi supervised 3D multimodal clinical prediction combines a small paired image and record set with a larger set of volumes without record annotations [7]. Paired cases align visual evidence with field semantics and clinical entities, while unpaired cases improve representation consistency without providing hidden text targets. Unlike semi supervised segmentation, which assigns voxel masks, this task predicts heterogeneous fields with both image grounded and partly unobservable content. Earlier STS studies demonstrate the value of unannotated dental CBCT [8], and U-Mamba2-SSL provides a staged combination of image pretraining, consistency learning, and pseudo label refinement [9]. Mean Teacher supplies stable temporal targets [10], but reliability controls remain necessary to prevent confirmation bias. Relevant strategies include translation agreement [11], selective mutual learning [12], uncertainty aware supervision [13], and voxel reliability constraints [14]. Clinical record prediction therefore requires these principles to act on cross modal representations and structured entities without corrupting the retrieval space learned from paired records.

We propose GQ-TSR, named Guarded Query Teacher Student Retrieval, which formulates the task as structured representation learning followed by fieldwise clinical retrieval. As illustrated in Fig. 1(a), image only masked volume pretraining uses all available training CBCT scans to initialize a shared three dimensional encoder. Fig. 1(b) shows GL3D-SQNet, which combines full volume context with overlapping local views and uses structured queries to produce field, record, and clinical entity representations. Supervised training aligns these outputs with compact targets derived from the paired records. The trained network then initializes the homogeneous student and EMA teacher in Fig. 1(c). During this stage, consistency between weak and strong views and selectively retained online and cross model entity pseudo labels allow the model to learn from unpaired scans. A model guard accepts the joint model only when it preserves supervised stability and improves retrieval behaviour. Otherwise, the supervised model is retained. At inference, each field is independently selected from a fixed bank of labeled records using semantic similarity, soft entity agreement, and reference density correction. Predictions from the training folds are finally combined to form the seven field record.

2 Method

2.1 Overview

Let $\mathcal{D}_l = \{(x_i, r_i)\}_{i=1}^{N_l}$ be the paired set and $\mathcal{D}_u = \{x_j^u\}_{j=1}^{N_u}$ the unpaired volumes. Record $r_i = \{r_i^f\}_{f\in\mathcal{F}}$ contains fields indexed by $\mathcal{F}$, and $\mathcal{M}$ indexes the clinical entity families. Given x_q, GQ-TSR predicts $\hat{r}_q$. Each training partition builds a fixed Reference Bank from $\mathcal{D}_l$, containing the original field texts and their semantic and entity targets. No sample from $\mathcal{D}_u$ enters the bank.

Figure 1 follows the training and inference path. Image based pretraining initializes GL3D-SQNet. Supervised learning aligns its query outputs with text

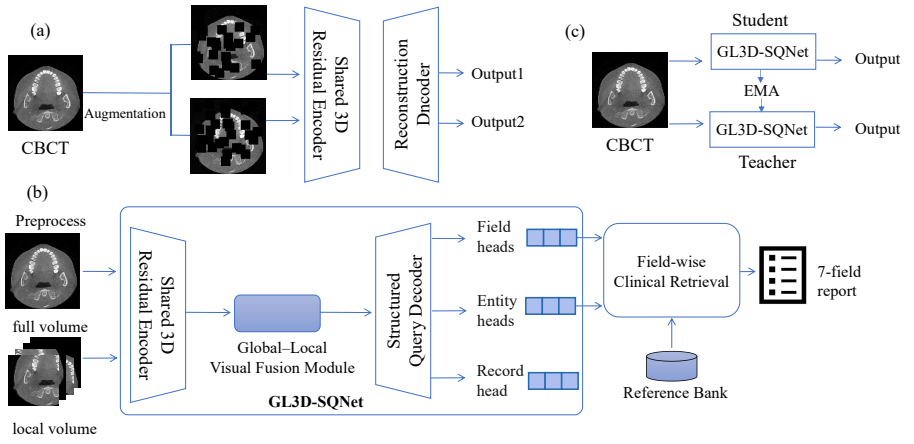


Fig. 1. GQ-TSR overview. (a) Masked volume pretraining initializes the shared encoder. (b) GL3D-SQNet fuses full and local features, then predicts field, entity, and record representations. Retrieval compares these outputs with the fixed Reference Bank. (c) The student and EMA teacher share GL3D-SQNet.

and entity targets. A student and EMA teacher then learn from unpaired volumes, subject to a model guard. At inference, the selected model retrieves each field from the Reference Bank.

2.2 CBCT Preprocessing and Image Based Pretraining

Each CBCT volume is reoriented, intensity normalized, cropped around an enlarged foreground box, and resampled to a common support, giving $\bar{x}$. Pretraining uses $\bar{x}$ from both data sets without clinical records. Independent appearance transforms a_k and block masks m_k produce two views

$$x_k = a_k(\bar{x}), \qquad v_k = (1 - m_k) \odot x_k, k \in \{1, 2\}, \tag{1}$$

where v_k is the masked view and $\odot$ denotes elementwise multiplication.

Let E_θ, R_ψ, and P_ϕ denote the shared encoder, temporary reconstruction decoder, and projection head. For $h_k = E_\theta(v_k)$, the decoder predicts $\hat{x} = R_\psi(h_1)$ and the projection head gives $u_k = \mathrm{norm}(P_\phi(h_k))$. With voxel index s, foreground weight $w(s)$, small constant ϵ, and stop gradient operator sg, the pretraining losses are

$$\mathcal{L}_{\mathrm{rec}} = \frac{\sum_s m_1(s) w(s) |\hat{x}(s) - \bar{x}(s)|}{\sum_s m_1(s) w(s) + \epsilon}, \tag{2}$$

$$\mathcal{L}_{\mathrm{view}} = \|u_1 - \mathrm{sg}[u_2]\|_2^2. \tag{3}$$

Reconstruction learns local anatomy, while view consistency reduces sensitivity to appearance changes. Only E_θ is transferred to GL3D-SQNet.

2.3 GL3D-SQNet and Cross Modal Supervision

For case i, the full volume $\bar{x}_i$ and local views $\mathcal{C}_i = \{c_{i,k}\}_{k=1}^{K_i}$ share the pretrained encoder. The fusion module Φ forms visual memory H_i. A decoder then applies record query q^r, field queries $\{q^f\}$, and entity queries $\{q^m\}$:

$$H_i = \Phi\left(E_\theta(\bar{x}_i), \{E_\theta(c_{i,k})\}_{k=1}^{K_i}\right), \tag{4}$$

$$U_i = \mathrm{Dec}(Q, H_i), \quad Q = \{q^r, \{q^f\}_{f\in\mathcal{F}}, \{q^m\}_{m\in\mathcal{M}}\}. \tag{5}$$

Prediction heads map U_i to the normalized record embedding z_i^r, field embeddings z_i^f, entity probabilities p_i^m, and count distributions π_i^m.

Field projection g_f and record projection g_r define fixed text targets within each training partition [15,16]:

$$t_i^f = \mathrm{norm}(g_f(r_i^f)), \qquad t_i^r = \mathrm{norm}(g_r(r_i)). \tag{6}$$

The record also yields entity target y_i^m and count target c_i^m. Supervised alignment and structured prediction use

$$\mathcal{L}_{\text{field}} = \frac{1}{|\mathcal{F}|}\sum_{f\in\mathcal{F}}[1-\cos(z_i^f, t_i^f)], \qquad \mathcal{L}_{\text{record}} = 1-\cos(z_i^r, t_i^r), \tag{7}$$

$$\mathcal{L}_{\text{entity}} = \sum_{m\in\mathcal{M}}\mathrm{BCE}(p_i^m, y_i^m), \qquad \mathcal{L}_{\text{count}} = \sum_{m\in\mathcal{M}}\mathrm{CE}(\pi_i^m, c_i^m). \tag{8}$$

Field and record contrastive losses supply labeled negatives. The original field texts, t_i^f, and y_i^m form the Reference Bank used at inference.

2.4 Guarded Teacher and Student Learning

The supervised model initializes a student and an identical teacher. At update s, the teacher parameters follow the student by EMA [10]:

$$\theta_T^{(s)} = \alpha_s\theta_T^{(s-1)} + (1-\alpha_s)\theta_S^{(s)}. \tag{9}$$

where, α_s is the EMA decay. For unpaired volume x_j^u, the teacher receives weak view $a_w(x_j^u)$ and the student strong view $a_s(x_j^u)$. Their field and record embeddings satisfy [17,18]

$$\mathcal{L}_{\text{cons}} = \frac{1}{|\mathcal{F}|}\sum_{f\in\mathcal{F}}\|z_{j,S}^f - \mathrm{sg}[z_{j,T}^f]\|_2^2 + \|z_{j,S}^r - \mathrm{sg}[z_{j,T}^r]\|_2^2, \tag{10}$$

where S and T identify the two branches. Teacher entity probability $p_{j,k}^{m,T}$ supplies a pseudo label only outside an uncertain interval:

$$\hat{y}_{j,k}^m = \begin{cases} 1, & p_{j,k}^{m,T} \geq \tau_m^+, \\ 0, & p_{j,k}^{m,T} \leq \tau_m^-, \\ \varnothing, & \text{otherwise}, \end{cases} \tag{11}$$

where, τ_m^+ and τ_m^- are confidence boundaries and $\varnothing$ marks an ignored target. Positive evidence is required, negatives are limited, and offline targets require agreement across supervised models. A variance constraint prevents embedding collapse. These losses are added gradually while the supervised loss remains active.

Let $G(\theta_J;\theta_W)$ denote the guard for joint candidate θ_J and supervised model θ_W. It requires better retrieval without losing diversity or supervised stability:

$$\theta^* = \begin{cases} \theta_J, & G(\theta_J;\theta_W) = 1, \\ \theta_W, & G(\theta_J;\theta_W) = 0. \end{cases} \tag{12}$$

2.5 Fieldwise Clinical Retrieval

Each field independently queries the fixed Reference Bank. Semantic similarity is corrected for references that are close to many queries:

$$s_f^{\mathrm{sem}}(q,i) = \cos(z_q^f, t_i^f), \tag{13}$$

$$h_f(i) = \frac{1}{|\mathcal{N}_H(i)|} \sum_{q' \in \mathcal{N}_H(i)} s_f^{\mathrm{sem}}(q', i),$$

$$\widetilde{s}_f^{\mathrm{sem}}(q,i) = s_f^{\mathrm{sem}}(q,i) - \eta_h h_f(i), \tag{14}$$

where $\mathcal{N}_H(i)$ contains the queries nearest to reference i and η_h controls the correction [19].

Soft entity agreement complements the semantic score:

$$e_m(q,i) = \frac{2(p_q^m)^\top y_i^m}{\|p_q^m\|_1 + \|y_i^m\|_1 + \epsilon}, \tag{15}$$

$$s_f^{\mathrm{ent}}(q,i) = \frac{\sum_{m \in \mathcal{M}_f} \omega_{f,m} e_m(q,i)}{\sum_{m \in \mathcal{M}_f} \omega_{f,m} + \epsilon},$$

$$S_f(q,i) = \widetilde{s}_f^{\mathrm{sem}}(q,i) + \eta_e s_f^{\mathrm{ent}}(q,i), \tag{16}$$

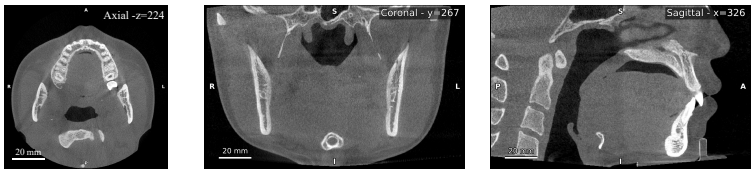
where, $\mathcal{M}_f$ contains the entity families relevant to field f, $\omega_{f,m}$ gives their relevance, and η_e controls reranking. Each model copies its highest scoring nonempty text. Exact text voting combines the models, with retrieval scores resolving ties. Fields may come from different references, but every output remains traceable to a labeled record.

3 Experiments

3.1 Dataset and Strictly Isolated Protocol

The MMDental Task 3 data contained 50 labeled CBCT volumes paired with seven field records, 200 unlabeled volumes, and 50 public validation volumes with hidden records [2]. Each validation case required all seven record fields

from CBCT alone. The qualitative prediction in Fig. 2 shows that GQ-TSR can generate a structurally complete seven-field record for an unseen CBCT volume. This example primarily demonstrates output completeness and field-level organization.



Field	Model inference text
Main appeal	Biting discomfort on the right back tooth for several weeks
Present medical history	The patient had a dental implant today and had preoperative scaling before coming to our hospital for treatment.
Oral Check	There are deep and large caries holes in the crown. When probing and percussing, the teeth are not loose. The teeth are sensitive and painful to cold stimulation. ...
Diagnosis	Acute pulpitis Impacted tooth Dental caries
Treatment plan	It is recommended to remove the restoration and perform root canal retreatment and crown restoration. Filling treatment. removal.
Handle	The patient was informed of the plan and agreed. The patient chose ITI pure titanium implants and made an appointment for surgery.
Doctor advices	Follow-up for discomfort.

Fig. 2. Qualitative prediction for public validation case 19. The top row shows axial, coronal, and sagittal CBCT views. The bottom table reports the seven fields inferred by GQ-TSR: Main appeal, Present medical history, Oral Check, Diagnosis, Treatment plan, Handle, and Doctor advices. Ground-truth text is not shown because the public validation records were hidden.

3.2 Implementation Details

Table 1 summarizes the training configuration.

CBCT intensities were clipped by robust percentiles, normalized to $[0, 1]$, cropped around the foreground, and resampled to $96 \times 160 \times 160$. The model processed the complete volume together with up to four overlapping local crops. Spatial flips were disabled because an unsynchronized flip changes the meaning of FDI tooth labels.

Image-only pre-training used all labeled and unlabeled CBCT volumes. The downstream stage used five folds; each fold fitted its text projections only on the fold-training records. The student was initialized with the pre-trained encoder, and the teacher was updated from the student by EMA. Supervised warm-up preceded joint training. In the joint stage, each update combined one labeled case

Table 1. Training protocol used by GQ-TSR.

Component	Setting
Input and local views	$96 \times 160 \times 160$ volume; up to four $64 \times 112 \times 112$ crops
Visual network	3D ResNet-34; hidden width 512; 2 visual and 4 query layers
Text targets	Character TF-IDF (2–5 grams), truncated SVD, 64 dimensions
Pre-training	20,000 steps; mask ratio 0.45; AdamW; peak LR 2×10^{-4}
Fold protocol	5 folds; at most 1,000 supervised and 1,000 joint steps per fold
Batch configuration	Batch size 1 per branch; gradient accumulation over 2 steps
Supervised learning rates	Encoder 1×10^{-5}; remaining modules 5×10^{-5}
Joint learning rates	Encoder 2×10^{-6}; remaining modules 1×10^{-5}
Optimization	Weight decay 0.02; cosine decay; AMP; gradient clipping at 1.0
EMA momentum	Scheduled from 0.990 to 0.9995
Model size	39,957,617 parameters per student or teacher

and one unlabeled case before gradient accumulation. AdamW, cosine learning-rate decay, automatic mixed precision, and gradient clipping were used throughout.

4 Results and Discussion

4.1 Challenge Results

GQ-TSR was evaluated on all 50 public validation cases and produced no missing predictions. It obtained a Weighted Score of 0.26. Table 2 reports the official text and clinical metrics.

Field completion reached 1.00, compared with 0.90 for the references. The clinical metrics show a recall bias, most clearly for tooth notation, where recall was 0.70 and precision was 0.09. Exact diagnosis match remained 0.00, and the low higher order text scores indicate limited phrase level agreement.

4.2 Discussion and Limitations

GQ-TSR produces complete records whose fields can be traced to labeled references. Structured queries separate field, record, and entity supervision, while the model guard rejects semi supervised updates that do not preserve held out retrieval quality. These choices reduce unsupported free text and limit damage from unreliable pseudo labels.

Retrieval from a closed reference bank also limits coverage. The model cannot express unseen diagnoses or treatments, and independent field selection may

Table 2. Official results on the 50-case public validation set.

Text and completion metrics			
Metric	Score	Metric	Score
BLEU-1	0.13	BLEU-2	0.04
BLEU-3	0.02	BLEU-4	0.01
METEOR	0.18	ROUGE-L	0.17
CIDEr	0.04	Prediction field completion	1.00
Structured clinical metrics			
Target	Precision	Recall	F1 / match
Tooth notation	0.09	0.70	0.15
Diagnosis code	0.05	0.12	0.07
Treatment action	0.27	0.47	0.32
Medication	0.64	0.64	0.64
Diagnosis exact / partial match	N/A	N/A	0.00 / 0.12
Treatment plan recall	N/A	0.53	N/A

combine statements from different patients. Reference hubness reduces text diversity, while the absence of tooth or lesion annotations weakens spatial grounding. The low tooth-notation precision and diagnosis agreement are consistent with this limitation.

The medication F1 of 0.64 should be interpreted cautiously because many medication fields are empty. Its internal F1 on nonempty references was 0.00. Unpaired CBCT improves visual consistency but cannot supply new clinical language. Future work should study case level retrieval, structured tooth supervision, and compact medical text representations under the same data isolation and model guard.

5 Conclusion

We introduced GQ-TSR for predicting seven clinical record fields from dental CBCT when paired records are limited. Image only pretraining first learns three dimensional structure from labeled and unlabeled scans. Global and local features are then read by field, record, and entity queries, while a guarded EMA teacher and student stage uses unlabeled data only when retrieval quality is preserved. Fieldwise retrieval keeps the final text traceable to labeled records and avoids unconstrained generation. On the official public validation set, GQ-TSR obtained a Weighted Score of 0.26, a field completion rate of 1.00, and no missing predictions. The results show reliable output coverage but limited diagnostic specificity, particularly for exact diagnosis and higher order text agreement. The closed reference bank and weak tooth level grounding remain the main constraints. Future work should improve spatial supervision and case level coherence without losing data isolation or reference provenance.

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